

Donghyun Lee

CONTACT INFORMATION Haedong Advanced Engineering Center **Phone:** (+82) 010-5787-4336
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EDUCATION *Seoul National University* Mar. 2022 - Present
M.S./Ph.D. Student in Computer Science and Engineering Seoul, Korea

- Advisor: Professor Jae W. Lee

Yonsei University Mar. 2018 - Feb. 2022
Bachelor of Science in Electrical and Electronic Engineering Seoul, Korea

- Awarded Highest Honors at Graduation (GPA: 4.14 / 4.30)

RESEARCH INTERESTS Efficient 3D Vision, Systems for Machine Learning, SW/HW Co-Design, GPU/TPU kernel optimization

PUBLICATIONS **QUESO: Storage-Assisted Quantization Error Compensation for On-Device LLM Inference**

Seong Hoon Seo, **Donghyun Lee**, Geonha Lee, Hojoon Kim, Yeonhong Park, and Jae W. Lee
The ACM SIGOPS Annual Technical Conference (**ATC**), 2026 (Under review)

GS-Scale: Unlocking Large-Scale 3D Gaussian Splatting Training via Host Offloading

Donghyun Lee, Dawoon Jeong, Jae W. Lee, and Hongil Yoon
The 31st ACM International Conference on Architectural Support for Programming Languages and Operating Systems (**ASPLOS**), Pittsburgh, PA, 2026

FastPoint: Accelerating 3D Point Cloud Model Inference via Sample Point Distance Prediction

Donghyun Lee, Dawoon Jeong, Jae W. Lee, and Hongil Yoon
IEEE/CVF International Conference on Computer Vision (**ICCV**), Honolulu, Hawaii, October 2025

FACIL: Flexible DRAM Address Mapping for SoC-PIM Cooperative On-device LLM Inference

Seong Hoon Seo, Junghoon Kim, **Donghyun Lee**, Seonah Yoo, Seokwon Moon, Yeonhong Park, and Jae W. Lee
The 31st IEEE International Symposium on High Performance Computer Architecture (**HPCA**), Las Vegas, NV, March 2025.

Frugal 3D Point Cloud Model Training via Progressive Near Point Filtering and Fused Aggregation

Donghyun Lee, Yejin Lee, Jae W. Lee, and Hongil Yoon
European Conference on Computer Vision (**ECCV**), Milan, Italy, September 2024.

Liquid: Mix-and-Match Multiple Image Formats to Balance DNN Training Pipeline

Woohyeon Baek*, Jonghyun Bae*, **Donghyun Lee**, Hyunwoong Bae, Yeonhong Park, and Jae W. Lee

14th ACM SIGOPS Asia-Pacific Workshop on Systems (**APSys**), Seoul, South Korea, August 2023.

Not All Neighbors Matter: Point Distribution-Aware Pruning for 3D Point Cloud

Yejin Lee, **Donghyun Lee**, JungUk Hong, Jae W. Lee, and Hongil Yoon

37th AAAI Conference on Artificial Intelligence (**AAAI**), Washington, DC, February 2023.

INTERNSHIPS

Google

Feb. 2026 - Present

Google Student Researcher (Collaborator: Hongil Yoon)

Seoul, Korea

- Performance modeling for Pallas TPU kernels and optimizing kernels for upcoming TPU architectures.

Architecture and Code Optimization Lab (ARC Lab)

Jul. - Aug. 2021

Undergraduate Student Intern

Seoul, Korea

- Implement block-sparse GEMM CUDA kernel for sparse DNN training

Compiler Research Laboratory (Corelab)

Apr. 2020 - Jun. 2021

Undergraduate Student Intern

Seoul, Korea

- Contributed to FlexC graph compiler project

HONORS AND AWARDS

The 31st Samsung Humantech Paper Award: Encouragement Prize, Samsung Electronics Co., Ltd.

2024 Accelerator Programming Winter School, CUDA competition, 1st place, SNU THUNDER Research Group & Manycoresoft

Highest Honors at Graduation (Top 1% of class), Yonsei University

TEACHING EXPERIENCE

Computer Architecture (Instructor: Prof. Jae W. Lee)

Seoul National University, Sep. - Dec. 2023 Teaching Assistant

- Undergraduate course taught in English

SERVICES

Conference Reviewer, 43rd International Conference on Machine Learning 2026, The 40th Annual Conference on Neural Information Processing Systems 2026

Student Volunteer, International Symposium on Code Generation and Optimization 2022

COMPUTER SKILLS

Laanguages: C/C++, CUDA, Verilog, Python

Applications/Frameworks: Pytorch, Tensorflow, MLX

REFERENCES

Jae W. Lee

Professor of Computer Science and Engineering at SNU

jaewlee@snu.ac.kr

Relationship: Ph.D. Advisor

Hongil Yoon

Technical Lead & Manager at Google

hongilyoon@google.com

Relationship: Academic Advisor